

Homework 4

Autonomous Mobile Robots

Sravanth Chowdary Potluri und5uv

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Instructions And Notes

This document provides the results of the codes for the assignment provided in the Codes folder. All the instructions to run the code are provided as comments in the code. Also please note that the simulation time in Q2 has been set to 10 seconds to give enough time for the diamond trajectory to complete. This can be changed back to 5, the default time in the simulation_2d.m file. The controller gains in both the questions have been tuned to accomplish the objectives satisfactorily.

Question 1

In this assignment, you will get started with a 1D quadrotor simulator and implement a controller for it. We will be testing your controller with two cases. In the first test case, the quadrotor simply needs to stabilize at a height of 0. The second test case gives the quadrotor a step input of 1 meter; that is, your quadrotor will be asked to rise to a height of 1 meter. The aim of this assignment is to get your controller to achieve a desired response for each test case.

Solution

Plots attached

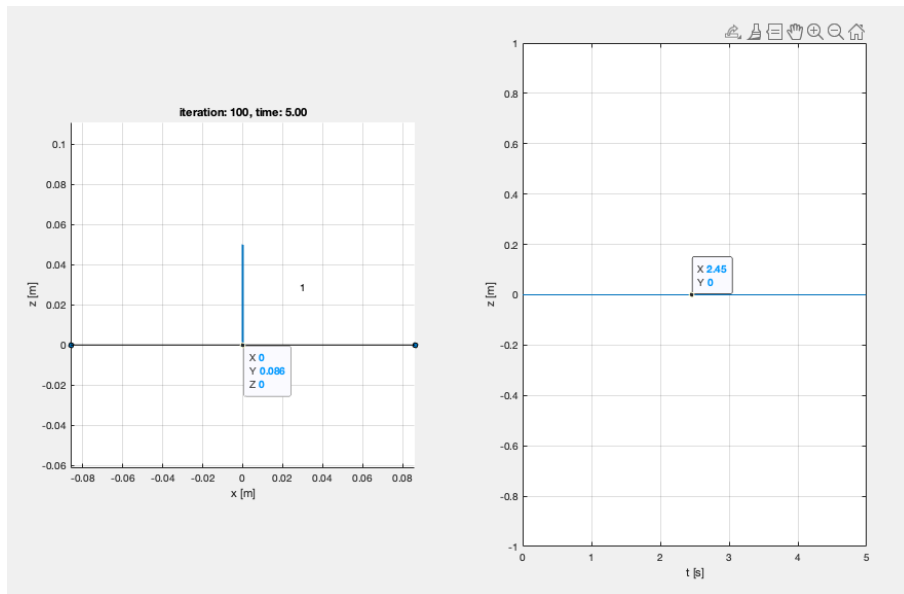


Figure 1: 1D Quadrotor Stabilize

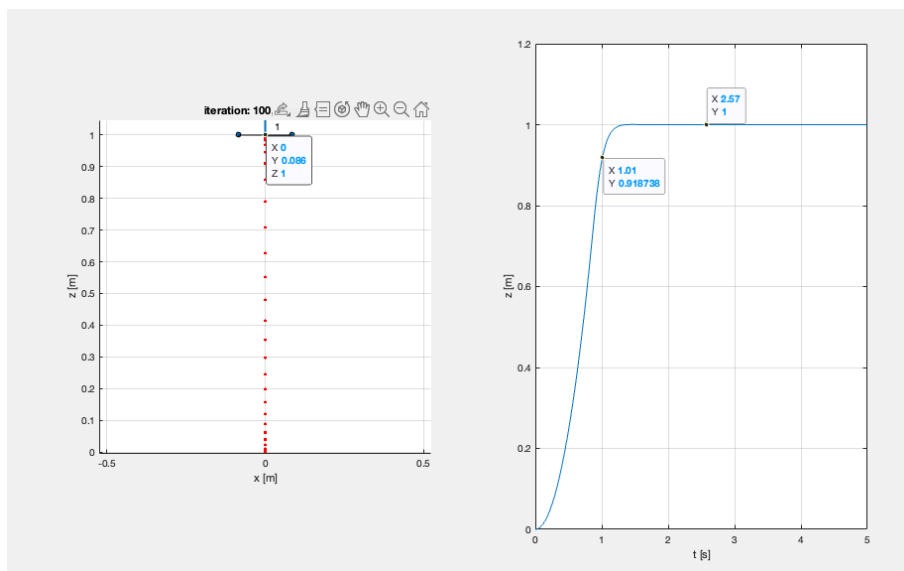


Figure 2: 1D Quadrotor Step Input 1

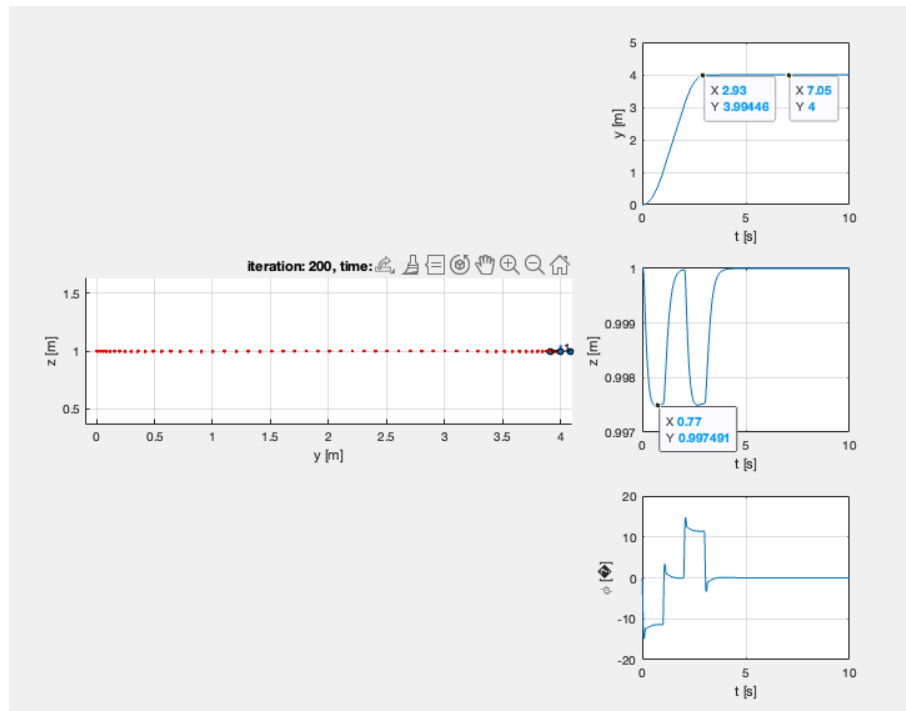


Figure 3: 2D Quadrotor Trajectory - Line

Question 2

In this assignment, you will design and implement a controller for a planar (2D) quadrotor. We will be testing your controller with two trajectories, the first one is a horizontal line trajectory and the second one is a more complex sine wave shaped trajectory. The aim is to get your controller to follow each trajectory such that the error between the desired position from the trajectory and the actual position of the robot is small.

Solution

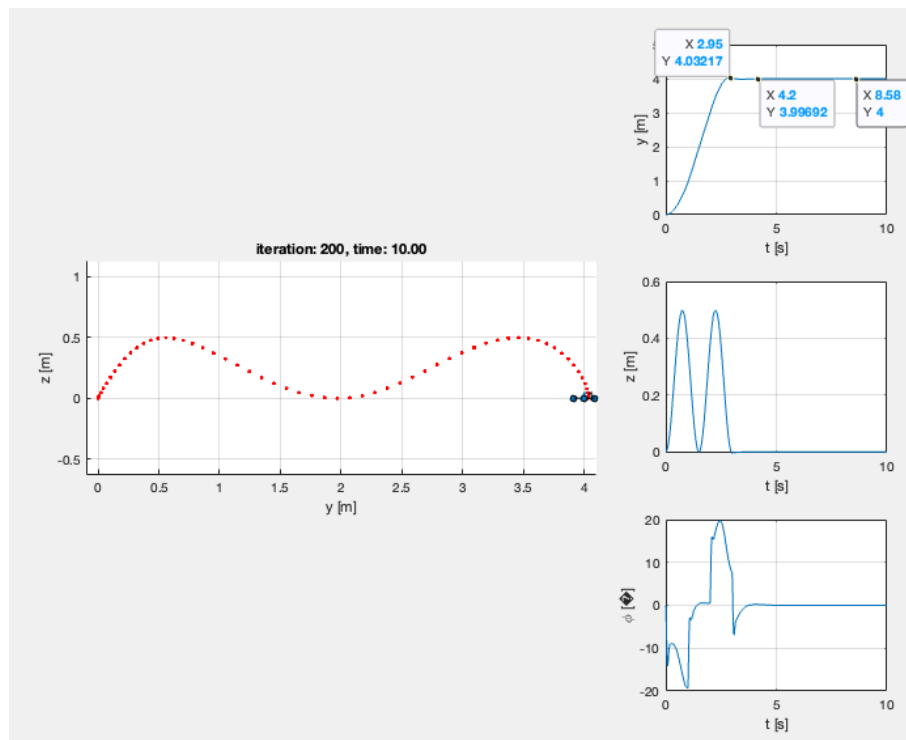


Figure 4: 2D Quadrotor Trajectory - Sine

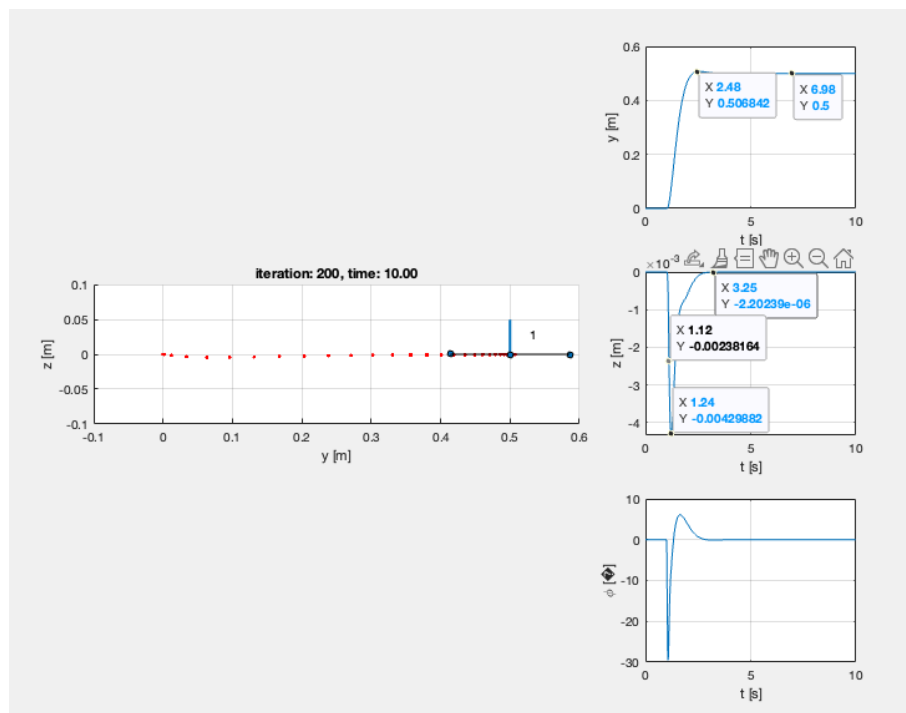


Figure 5: 2D Quadrotor Trajectory - Step

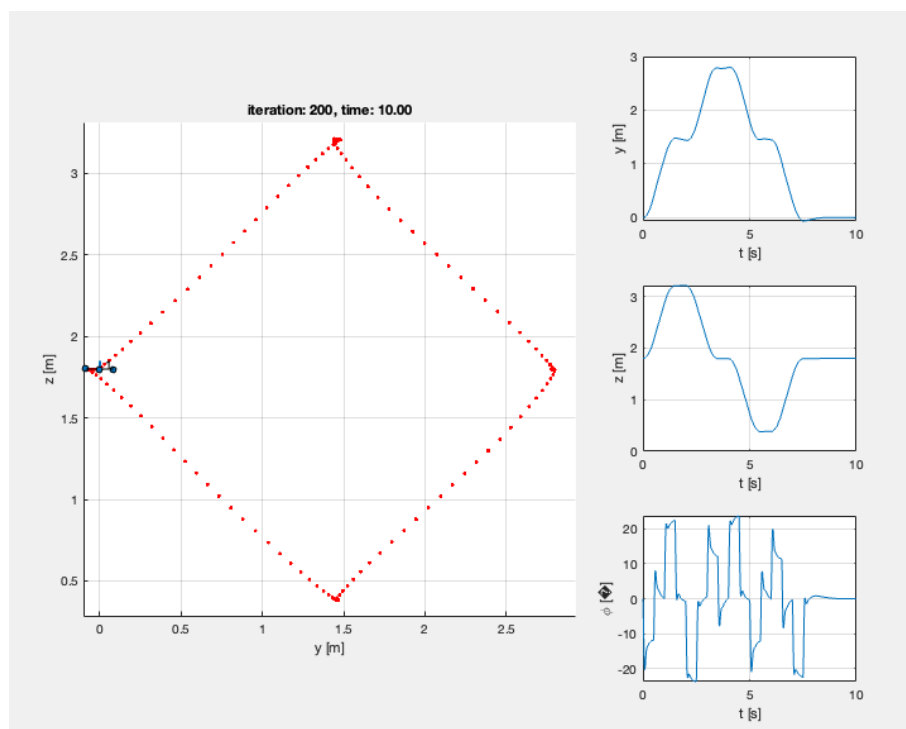


Figure 6: 2D Quadrotor Trajectory - Diamond